PGx Pipeline Presentation

PGx Pipeline: Strategy & Feasibility Presentation

Pharmacogenetics Integration for Clinical NGS Platform

20-Minute Product Team Presentation

Agenda

1. Current State of Pharmacogenomics (5 min)
2. Architectural Defense (7 min)
3. Feasibility Analysis (5 min)
4. Product & UI/UX Vision (3 min)

Part 1: Current State of the Art

1.1 Industry Approaches to Pharmacogenomics

Leading Approaches

Approach	Examples	Coverage
Targeted PGx Panels	Genotyping arrays	15-30 genes, SNPs only
WES/WGS-derived PGx	PharmCAT, Aldy	Broader coverage, variable quality
Clinical Decision Support	EHR-integrated alerts	Real-time prescribing guidance

Standard Reporting Elements

- Star-allele diplotypes (e.g., CYP2C19 *1/*17)
- Metabolizer phenotypes (Poor → Ultrarapid)
- CPIC/DPWG guideline-based interpretations
- Drug-specific dosing recommendations

1.2 PGx in Rare Disease Patients

Current Adoption Status

Status	Reality
Standard practice?	**No** - but rapidly emerging
Clinical value?	**High** - innovative and actionable
Implementation	Translational/precision medicine programs

Why Rare Disease Patients Need PGx

- **High medication burden** → More drug interactions
- **Polypharmacy** → Higher ADR risk
- **WES/WGS already available** → Reuse existing data
- **No additional testing required** → Cost-effective

> **Opportunity**: Convert diagnostic sequencing into therapeutic guidance

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1.3 Sequencing Modalities Comparison

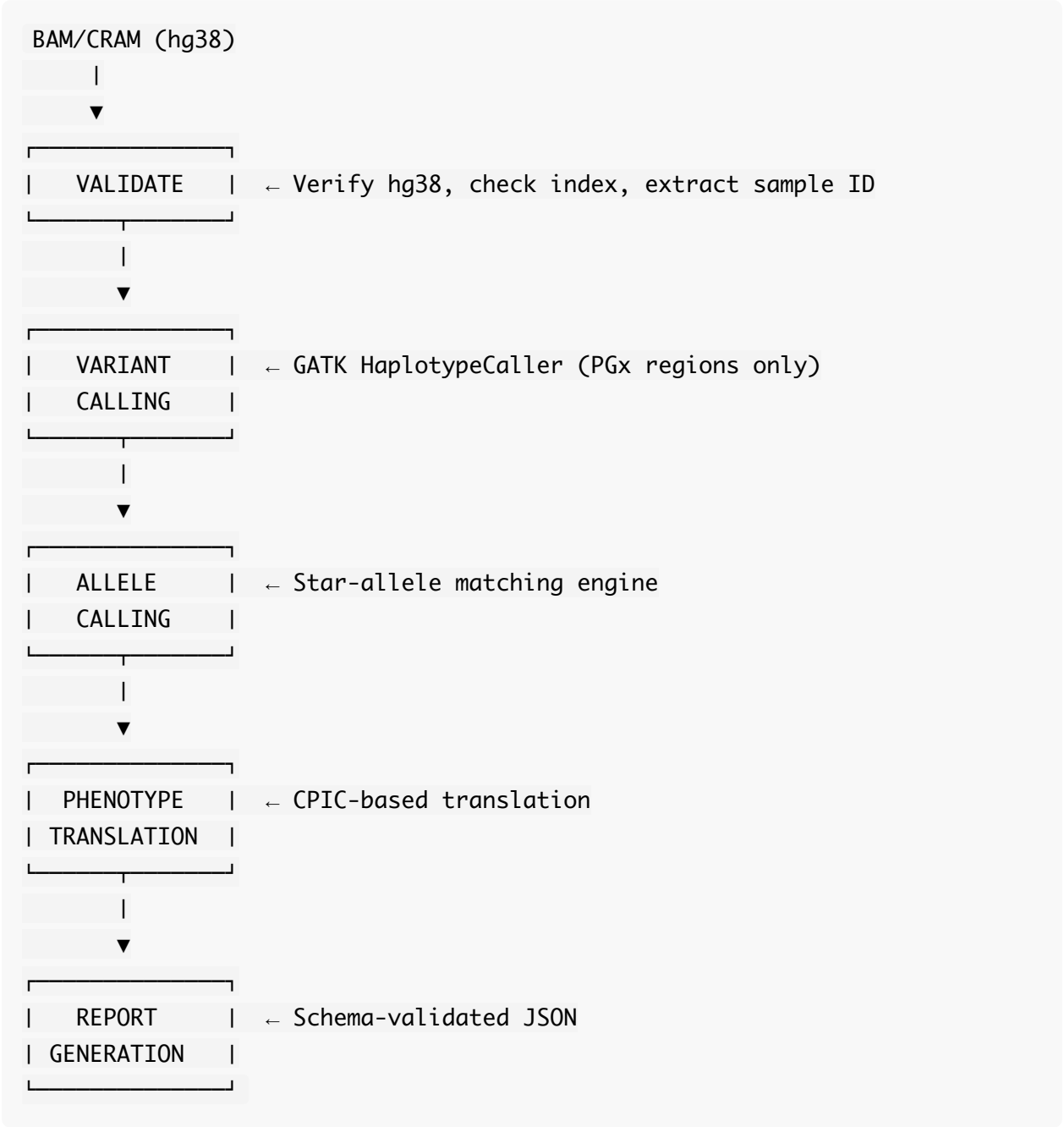
Modality	Pros	Cons	PGx Suitability
WES	Lower cost, widely available	Uneven coverage, misses regulatory regions	Medium
WGS	Uniform coverage, CNV detection	Higher cost, more data	High
Short-read	Industry standard, validated	Complex loci difficult (CYP2D6)	High (most genes)
Long-read	Superior for structural variants	Limited availability, cost	Future for complex genes

Industry Trajectory

- **Near-term**: Short-read WGS as clinical standard
- **Future**: Long-read for complex PGx genes (CYP2D6)

Part 2: Architectural Defense

2.1 Pipeline Architecture Overview



2.2 Technology Stack Rationale

Component	Choice	Why
Orchestration	Nextflow DSL2	Industry standard, reproducible, scalable
Variant Calling	GATK HaplotypeCaller	Clinical-grade, well-validated
Logic Layer	Python 3.11+	Type safety, Pydantic models, testability
Containers	Docker	Reproducibility, version control
Data Models	Pydantic v2	Validation, serialization, type hints
Reference Data	CPIC/PharmVar	Regulatory-compliant, peer-reviewed

Key Design Principles

1. **Separation of concerns**: Heavy computation ≠ interpretation logic
2. **Deterministic outputs**: Pinned versions, no `:latest` tags
3. **Auditable**: Full provenance in output JSON
4. **Extensible**: Module contracts enable staged rollout

2.3 Algorithm Choices

Allele Calling Strategy

Mode A: VCF-based (Group 1 Genes)

```
VCF → Match variants → Identify haplotypes → Resolve diplotype
```

- Works for: CYP2C19, CYP2C9, VKORC1, SLCO1B1, TPMT, DPYD, UGT1A1, NUDT15

- Confidence: High (SNP-based)

Mode B: BAM-based (Future - Group 2)

```
BAM → CNV detection → Hybrid detection → Complex diplotype
```

- Required for: CYP2D6
- Tools: Cyrius, Stargazer, Aldy

Phenotype Translation

```
Diplotype → Activity Score → Phenotype → Prescribing Implication
    ↓             ↓              ↓                ↓
  *1/*17         2.5          Rapid          "Consider alternative"
```

2.4 Handling Complex PGx Genes

CYP2D6 Challenges

Challenge	Impact	Our Strategy
Gene deletions (*5)	Missed calls	BAM-based detection (v1.1)
Duplications (*1xN)	Copy number	Dedicated CNV module
CYP2D7 hybrids	False positives	Specialized caller integration
Pseudogene homology	Mismapping	Long-read future pathway

Confidence Scoring Formula

```
Confidence = 0.40 × Depth + 0.35 × Quality + 0.25 × Completeness
```

Quality Flags

- `LOW_DEPTH` - Any position < 10x
- `LOW_QUALITY` - QUAL < 20
- `INCOMPLETE_ALLELE` - Missing defining variants
- `AMBIGUOUS_CALL` - Multiple equally likely diplotypes

Part 3: Feasibility Analysis

3.1 Technical Limitations

NGS Limitations for PGx

Limitation	Affected Genes	Mitigation
WES coverage gaps	All	Use WGS, flag low-coverage
Short-read CNV detection	CYP2D6	Dedicated tools, defer to v1.1
Homologous region mapping	CYP2D6, HLA	Specialized callers
Regulatory region variants	CYP2C19 *17	WGS preferred

Group 1 Genes: High Feasibility

- All 8 genes are SNP-based
- Well-defined star alleles
- CPIC Level A evidence
- Standard variant calling sufficient

Group 2+ Genes: Requires Specialized Handling

- CYP2D6: CNV calling module
- HLA: Dedicated HLA typers

3.2 Clinical Validation Strategy

Phase 1: Pre-Release Validation

Validation Type	Method	Target
Reference samples	GeT-RM, Coriell	Known diplotypes
Orthogonal comparison	vs. targeted PGx panels	Concordance rate
CPIC concordance	Expected phenotypes	100% for Group 1

Phase 2: Gradual Rollout

```
Week 1-4:   Group 1 genes only (8 genes)
Week 5-8:   Add CYP2D6 with confidence flags
Week 9+:    HLA module (opt-in)
```

Documentation Requirements

- Performance characteristics per gene
- Known limitations and edge cases
- Confidence thresholds for clinical use
- Recommendations for confirmatory testing

3.3 Accuracy Expectations

Group 1 Genes (v1.0)

Gene	Expected Concordance	Confidence
CYP2C19	>99%	High
CYP2C9	>99%	High
VKORC1	>99%	High
SLCO1B1	>99%	High
TPMT	>98%	High
DPYD	>95%	Medium (rare variants)
UGT1A1	>95%	Medium (*28 is TA-repeat)
NUDT15	>98%	High

Part 4: Product & UI/UX Vision

4.1 Clinical Use Case

Scenario: Clopidogrel Prescription

```
┌─────────────────────────────────────────────────────────┐
│  PRESCRIBING: Clopidogrel 75mg                          │
├─────────────────────────────────────────────────────────┤
│  ⚠️  PGx ALERT: CYP2C19 *1/*2 (Intermediate Metabolizer)│
│                                                         │
│  Recommendation: Consider prasugrel or ticagrelor       │
│  Guideline: CPIC 2022                                   │
│                                                         │
│  [View Details]  [Override with Reason]  [Accept]       │
└─────────────────────────────────────────────────────────┘
```

Integration Points

- EHR prescribing workflow
- Clinical decision support alerts
- Patient summary dashboard

4.2 Communicating Uncertainty

Design Principles

1. **Don't overwhelm** - Hide raw genomic data by default
2. **Surface uncertainty clearly** - Confidence indicators
3. **Enable drill-down** - Expandable evidence sections

UI Confidence Indicators

Confidence	Display	Meaning
High (>0.9)	✅ Green	Report as-is
Moderate (0.7-0.9)	🟡 Yellow	Clinical review suggested
Low (<0.7)	🔴 Red	Confirmatory testing recommended

Example Report Card

```
┌─────────────────────────────────────────────┐
│  CYP2C19                                    │
│  ─────────────────────────────────────────  │
│  Diplotype:    *1/*17                       │
│  Phenotype:    Rapid Metabolizer      ✅     │
│  Confidence:   95%                          │
│                                             │
│  ▸ View variants (1 found)                  │
│  ▸ View guideline (CPIC 2022)               │
│  ▸ Drug implications                        │
└─────────────────────────────────────────────┘
```

4.3 Report Layering

Audience-Appropriate Views

Audience	View	Content
Physician	Summary	Phenotype + drug guidance
Pharmacist	Detailed	Activity scores + dosing
Lab Director	Technical	Variants + QC metrics
EHR System	Machine-readable	Full JSON

Summary

What We've Demonstrated

Area	Evidence
Industry knowledge	Current PGx landscape understanding
Technical strategy	Defensible architecture with clear rationale
Feasibility awareness	Honest assessment of limitations
Product vision	Clinician-centered UI/UX concepts

Key Differentiators

1. **Reuses diagnostic WGS** - No additional testing
2. **CPIC-compliant** - Regulatory defensibility
3. **Confidence scoring** - Transparent uncertainty
4. **Modular design** - Staged rollout capability
5. **Clinical-grade** - Schema validation, provenance

Questions?

Appendix Available

- Detailed technical rationale document
- Full pipeline specification
- Test strategy documentation
- Data model schemas